

COURSE MATERIAL	
SUBJECT	BASIC MECHANICAL ENGINEERING (ME23AES101)
UNIT	2
COURSE	B.TECH
SEMESTER	1-1
DEPARTMENT	MECHANICAL ENGINEERING
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1. COURSE OBJECTIVES

The students after completing the course are expected to

1. Get familiarized with the scope and importance of Mechanical Engineering in different sectors and industries.
2. Explain different engineering materials and different manufacturing processes.
3. Provide an overview of different thermal and mechanical transmission systems and introduce basics of robotics and its applications.

2. PREREQUISITES

Students should have knowledge on

1. Engineering Physics

3. SYLLABUS

UNIT II

Manufacturing Processes: Principles of Casting, Forming, Joining processes, Machining, Introduction to CNC machines, 3D printing, and Smart manufacturing.

Thermal Engineering – Working principle of Boilers, Otto cycle, Diesel cycle, Refrigeration and air-conditioning cycles, IC engines, 2-Stroke and 4-Stroke engines, SI/CI Engines, Components of Electric and Hybrid Vehicles.

4. COURSE OUTCOMES

1. **Understand** the different manufacturing processes (L1).
2. **Explain** the basics of thermal engineering and its applications (L2).
3. **Describe** the working of different mechanical power transmission systems and power plants and describe the basics of robotics and its applications (L2).

5. Co-PO / PSO Mapping

CAD/ CAD	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3												3	
CO2	3						1						3	
CO3	3												3	

6. LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Principles of Casting	T1
2		Forming, Joining processes	T1
3		Machining, Introduction to CNC machines	T1
4		3D printing, and Smart manufacturing.	T1, R1
5	2	Thermal Engineering – Working principle of Boilers,	R1
6		Otto cycle, Diesel cycle,	T1, R1
7		Refrigeration and air-conditioning cycles	T1, R1
8		IC engines, 2-Stroke and 4-Stroke engines	T1, R1
9	3	SI/CI Engines	T1, R1
10		Components of Electric and Hybrid Vehicles.	T1, R1
11		Discussion of Previous year university questions in question papers	T1, R1

7. ACTIVITY BASED LEARNING

- Hands on sand casting simulation

8. LECTURE NOTES

1.1 MANUFACTURING PROCESS: PRINCIPLES OF CASTING

- Casting is a manufacturing process where molten material (metal, plastic, or other substances) is poured into a mould cavity, which takes the shape of the desired object. Once the material solidifies, the mould is removed, leaving behind the finished product.

Principles of Casting**Mold Preparation:**

- Mold Material: The choice of mould material (sand, metal, ceramic) depends on the cast material and the desired product complexity.
- Mold Design: The mould must be designed to replicate the desired shape accurately and have appropriate venting and gating systems.
- Mold Cavity: The mould cavity is the space within the mould where the molten material will solidify.
- Molten Material Preparation:

- **Heating:** The material is melted to a specific temperature, ensuring it has the correct fluidity for filling the mould.
- **Alloying:** In metal casting, alloys may be added to improve strength, durability, or corrosion resistance.

Pouring:

- **Gating System:** The molten material is poured into the mould through a gating system that distributes it evenly.
- **Risers:** Risers are additional cavities in the mould that compensate for shrinkage during solidification.
- **Solidification:**
- **Cooling:** The molten material cools and solidifies, taking the shape of the mould cavity.
- **Shrinkage:** As the material cools, it shrinks, which can cause defects if not accounted for.

Mold Removal:

- **Breaking:** The mould is broken or removed to extract the finished product.
- **Finishing:** The product may require additional finishing processes like machining, polishing, or coating.

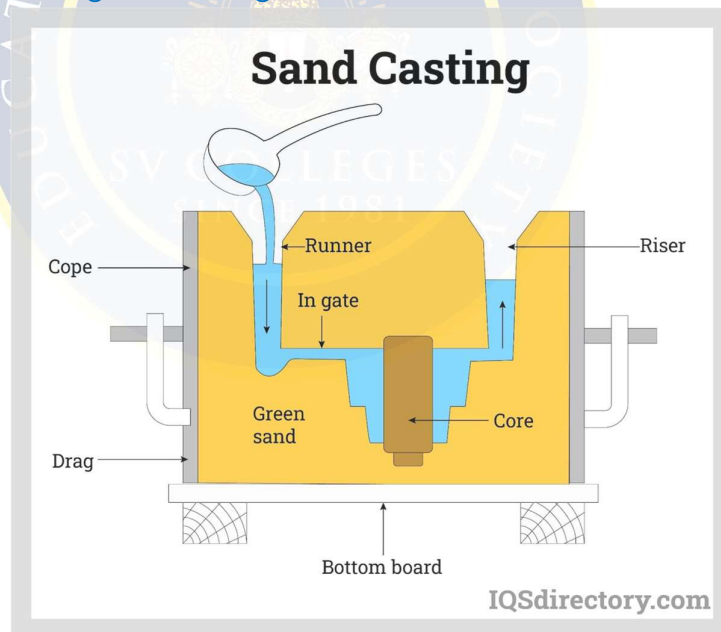


FIG. 2.1 Schematic representation of Sand Casting

1.2 FORMING

- Forming operates on the materials science principle of plastic deformation, where the physical shape of a material is permanently deformed.
- Forging*: Compressive forces shape a workpiece by hammering or pressing.
- Stamping*: Pressing a sheet of metal into a die to create specific shapes and features.
- Rolling*: Reducing thickness or changing cross-sectional areas of a workpiece by passing it through rolls.

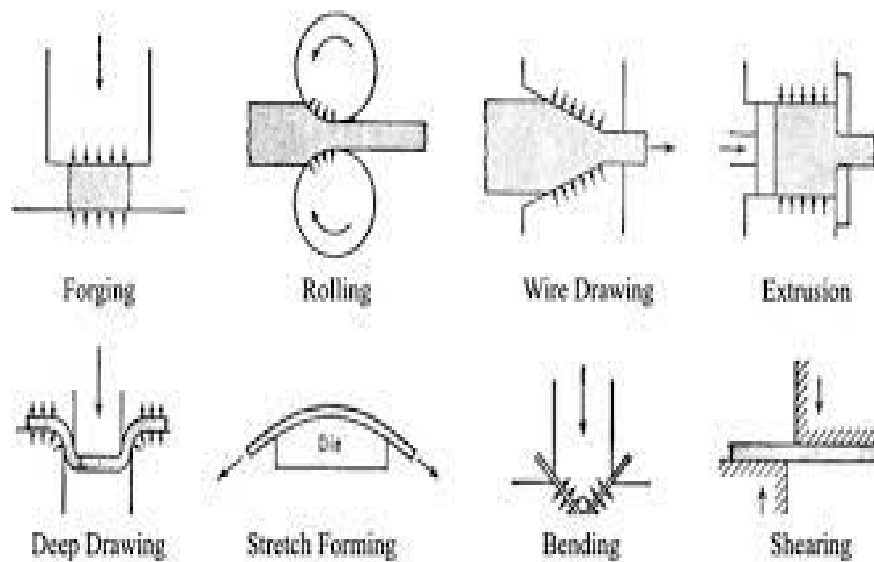


FIG. 2.2 Various forming process

1.3 JOINING PROCESSES

- Joining processes are techniques used to connect or combine different materials, components, or parts into a single unit. These processes are essential in various industries, including manufacturing, construction, and repair.

Types of Joining Processes

- Joining processes can be broadly classified into two categories:

1. Mechanical Joining:

- Fastening**: This involves using mechanical elements to hold parts together.
- Bolts and nuts**: The most common method, using threaded fasteners to connect parts.
- Screws**: Similar to bolts, but often have a head that prevents rotation.
- Rivets**: Permanent fasteners that are expanded after insertion.

- Pins and keys: Used to align and position parts.
- Press-fitting: Parts are forced together due to interference fits, creating a tight connection.
- Clamping: Parts are held together using clamps or other devices.
- Welding: Using heat and pressure to melt the edges of two materials and fuse them together.

2. Non-Mechanical Joining:

- Adhesives: Using a bonding agent to join materials.
- Glues: Used for joining materials like wood, paper, and plastics.
- Epoxy resins: Strong adhesives suitable for metals, ceramics, and composites.
- Soldering: Using a low-melting-point alloy to join metals, often with a flux.
- Brazing: Similar to soldering, but using a higher-melting-point alloy.
- Diffusion bonding: Joining materials at a high temperature and pressure, causing atoms to diffuse across the interface.
- Ultrasonic welding: Using high-frequency vibrations to create a bond between materials.
- Factors to Consider in Choosing a Joining Process:
- Material properties: The type of materials being joined.
- Required strength and durability: The load and environmental conditions the joint must withstand.
- Cost: The cost of the joining process and materials.
- Aesthetics: The appearance of the joint.
- Production speed: The time required to complete the joining process.
- Joint integrity: The reliability and longevity of the joint.

1.4 MACHINING

cutting tool removes metal from a rotating work piece to reduce its diameter; (b) drilling, in which a rotating drill bit is fed into the work to create a round hole; and (c) milling, in which a work part is fed past a rotating cutter with multiple edges. (d)Grinding: Abrasive grains are used to remove material and achieve high surface quality and tight tolerances.

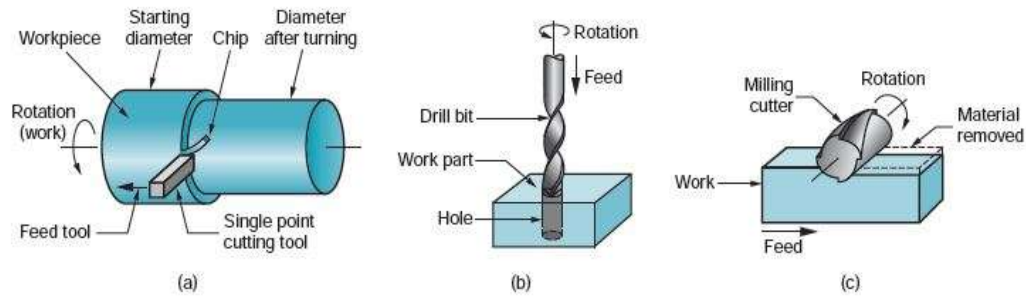


FIG. 2.3 Various Machining process

1.5 INTRODUCTION TO CNC MACHINES

CNC (Computer Numerical Control) machines are automated manufacturing devices that utilize computer programming and digital controls to perform various machining operations with a high level of precision, efficiency, and accuracy. These machines are widely used in the manufacturing industry to produce complex parts and components for various applications. Here's an introduction to CNC machines:

Key Components:

Computer Control Unit (CCU): The brain of the CNC machine that interprets the part program, processes the data, and controls the movement and actions of the machine.

Machine Control Unit (MCU): Controls the movement of the machine's components, such as the motors, drives, and spindle.

Input Devices: Keyboards, touchscreens, or other input devices are used to input the part program and other instructions.

Axes: CNC machines can have multiple axes (e.g., X, Y, Z) that define the directions in which the cutting tool can move.

Working Principle:

The CNC machine reads a part program, typically written in G-code, which contains a set of instructions for tool paths, tool speeds, tool changes, and other machining parameters.

The CNC controller interprets these instructions and converts them into signals to control the movement and actions of the machine's components. The cutting tool (e.g., end mill, drill bit) attached to the machine's spindle removes material from the workpiece based on the instructions in the part program.

Advantages of CNC Machines:

Precision and Accuracy: CNC machines offer high precision and accuracy, resulting in consistent and tight tolerances in the finished parts.

Efficiency and Productivity: CNC machines can operate continuously, reducing idle time and significantly improving production efficiency.

Flexibility and Versatility: CNC machines can be programmed to produce a wide range of parts and components, making them highly versatile for various applications.

Complex Geometries: They can create intricate and complex shapes that would be difficult or impossible to achieve manually.

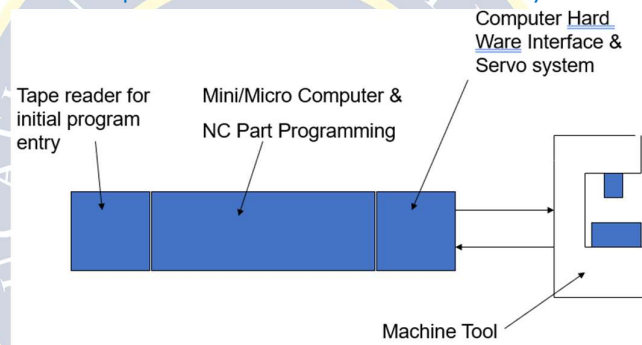


Fig 1.4 Schematic Diagram of CNC Machine

1.6 3D PRINTING

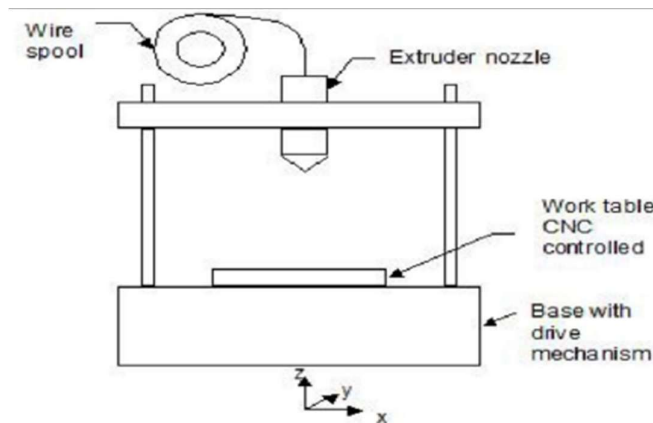


Fig 1.5 Schematic Diagram of 3D Printing

3D printing, also known as additive manufacturing, is a revolutionary technology that involves creating three-dimensional objects by layering materials based on a digital model.

Working Principle:

3D printing builds objects layer by layer, starting from the bottom and adding successive layers until the final 3D object is complete.

The process begins with a 3D digital model created using computer-aided design (CAD) software.

Types of 3D Printing Technologies:

1. Fused Deposition Modeling (FDM): Melts and extrudes thermoplastic materials through a nozzle to create layers. Commonly used for prototyping and creating functional parts.
2. Stereolithography (SLA): Uses a UV laser to cure liquid resin layer by layer, resulting in highly detailed and accurate parts. Commonly used in industries like jewelry, dental, and product design.
3. Selective Laser Sintering (SLS): Uses a laser to sinter powdered materials (e.g., plastics, metals) layer by layer. Suitable for producing functional prototypes and end-use parts.

Advantages:

Rapid Prototyping: Allows for rapid iteration and testing of designs, reducing development time.

Customization: Enables the creation of highly customized and individualized products.

Applications:

aerospace, automotive, healthcare, architecture, and consumer goods. It's used for rapid prototyping, custom manufacturing, and creating complex geometries that are challenging with traditional manufacturing methods.

1.7 SMART MANUFACTURING

Smart manufacturing is a technological approach that leverages digital technologies to optimize and improve manufacturing processes. It involves the integration of various technologies, such as the Internet of Things (IoT), artificial intelligence (AI), big data analytics, and automation, to create a connected and intelligent manufacturing ecosystem.

Key benefits of smart manufacturing include:

Increased efficiency through automation and data-driven decision-making.

Improved quality due to real-time monitoring and analysis of production data.

Enhanced flexibility to adapt to changing market demands and product designs.

Reduced costs through optimized resource utilization and waste reduction.

Improved sustainability by minimizing energy consumption and environmental impact.

Examples of smart manufacturing technologies:

IoT sensors for real-time data collection.

AI for predictive maintenance and quality control.

Robotics and automation for efficient production processes.

3D printing for rapid prototyping and customization.

1.8 WORKING PRINCIPLES OF BOILERS

Working Principle of a Boiler

A boiler is a device that generates steam or hot water by heating a working fluid (usually water) under pressure. The working principle of a boiler involves the following steps:

Water Heating: Water is fed into the boiler and heated to a high temperature.

Vaporization: As the water heats up, it reaches its boiling point and begins to vaporize, forming steam.

Pressure Increase: The steam is contained within the boiler, increasing its pressure.

Steam Generation: The high-pressure steam is generated and can be used for various applications, such as powering turbines, heating buildings, or industrial processes.

Classification of Boilers

Boilers can be classified based on various criteria, including:

Based on the Method of Heat Transfer:

Fire Tube Boilers: In these boilers, the hot gases from the combustion process pass through tubes surrounded by water. The heat is transferred from the hot gases to the water, causing it to boil.

Water Tube Boilers: In these boilers, water flows through tubes surrounded by hot gases. The heat is transferred from the hot gases to the water, causing it to boil.

Based on the Pressure of the Steam:

Low-Pressure Boilers: These boilers generate steam at a pressure of up to 10 bar. They are commonly used for heating purposes.

Medium-Pressure Boilers: These boilers generate steam at a pressure of 10 to 30 bar. They are used for a variety of industrial applications, such as powering turbines.

High-Pressure Boilers: These boilers generate steam at a pressure of 30 bar or more. They are used in power plants to generate electricity.

Based on the Fuel Used:

Coal-Fired Boilers: These boilers use coal as fuel to generate heat.

Oil-Fired Boilers: These boilers use oil as fuel to generate heat.

Gas-Fired Boilers: These boilers use natural gas or other gases as fuel to generate heat.

Electric Boilers: These boilers use electricity to heat the water.

Based on the Type of Water Used:

Steam Boilers: These boilers generate steam.

Hot Water Boilers: These boilers generate hot water.

1.9 OTTO CYCLE

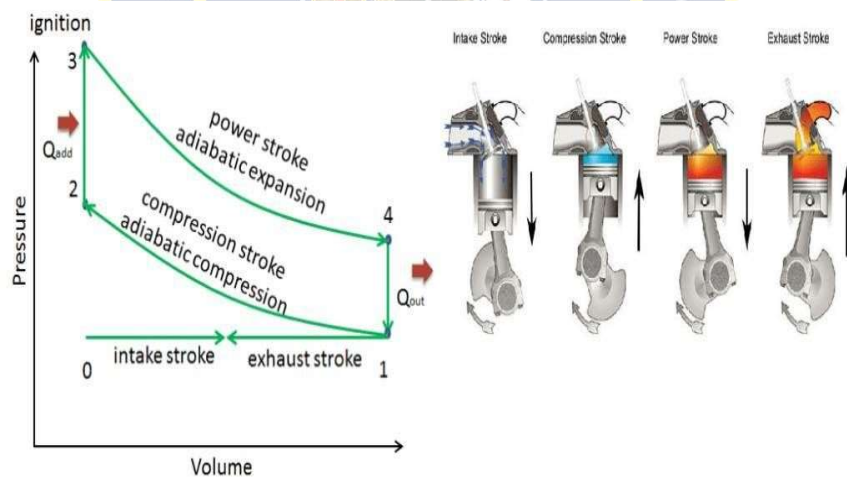


Fig 1.6 P-V Diagram of Otto cycle

1-2 (Adiabatic process): In this process compression takes place, as the piston moves from BDC to TDC increasing its temperature

2-3 (Isochoric process): In this process, ignition is taking place, combustion happens when the piston is at TDC and pressure increases at a constant volume.

3-4 (Adiabatic process): In this process expansion is taking place, the heat produced due to the combustion pushes the piston down which rotates the crankshaft.

4-1 (Isochoric process): In this process, heat rejection is taking place at constant volume. The compression ratio of the Otto cycle is 8 to 12.

1.10 DIESEL CYCLE

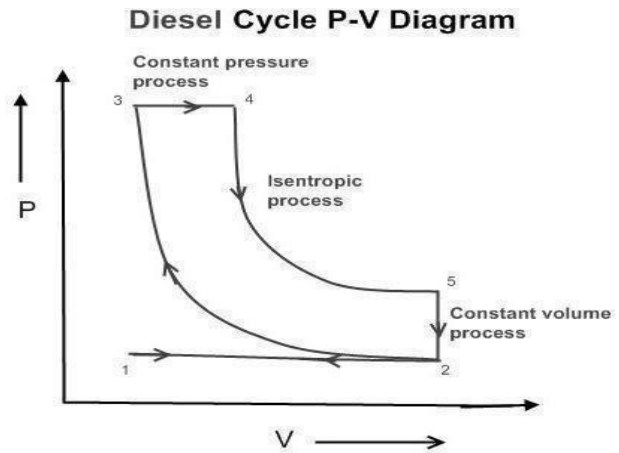


Fig 1.7 P-V Diagram of Diesel cycle

1-2 In this process suction stroke will take place.

2-3 (Adiabatic process) In this process compression takes place. Both the inlet and exhaust valves are closed and the compression takes place which is much higher than that of an Otto cycle. This increases the pressure and temperature.

3-4 (Isobaric process) In this process, fuel is added, and combustion occurs due to high temperature, while maintaining a constant pressure because the volume is also increasing.

4-5 (Adiabatic process) In this process expansion takes place, due to combustion the piston moves from TDC to BDC and power is generated.

5-2 (Isochoric process) In this process, heat rejection is taking place at constant volume.

Compression ratio is 14 to 22

1.11 REFRIGERATION AND AIR CONDITIONING CYCLES

Refrigeration and Air Conditioning

Refrigeration and air conditioning are both processes that involve removing heat from a space or object. While they share a common principle, they serve different purposes:

Refrigeration: The process of removing heat from a space or object to lower its temperature below the ambient temperature. This is used to preserve food, create cold storage, and produce ice.

Air Conditioning: The process of controlling the temperature, humidity, and air quality within a space to create a comfortable environment. This is used in buildings, vehicles, and industrial processes.

Working Principle of a Refrigeration System

A refrigeration system works on the principle of heat transfer and the properties of a refrigerant. The basic components of a refrigeration system include:

Compressor: Compresses the refrigerant gas, increasing its pressure and temperature.

Condenser: Cools the high-pressure, high-temperature refrigerant gas, causing it to condense into a liquid.

Expansion Valve: Reduces the pressure of the refrigerant liquid, causing it to evaporate and absorb heat from the surrounding environment.

Evaporator: The refrigerant, now a cold gas, absorbs heat from the space to be cooled, causing it to evaporate back into a gas.

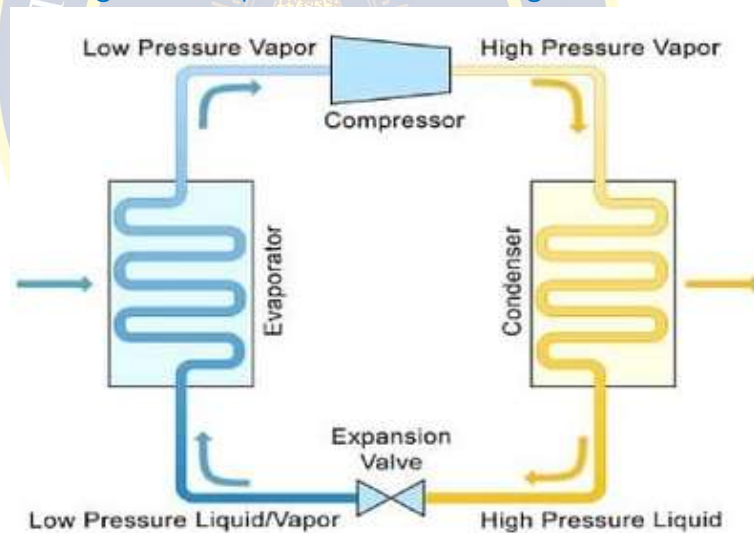


Fig 1.8 Refrigeration cycle

How it works:

Compressor: The compressor draws refrigerant gas from the evaporator and compresses it, increasing its pressure and temperature.

Condenser: The compressed refrigerant gas flows through the condenser, where it is cooled by a cooling medium (e.g., air or water). This causes the gas to condense into a liquid.

Expansion Valve: The liquid refrigerant flows through the expansion valve, which reduces its pressure. This causes the liquid to flash boil, absorbing heat from the surrounding environment.

Evaporator: The cold refrigerant gas flows through the evaporator, where it absorbs heat from the space to be cooled. This causes the refrigerant to evaporate back into a gas, completing the cycle.

1.12 IC ENGINES

An **Internal Combustion (IC) Engine** is a type of heat engine where the combustion of fuel occurs **inside the engine cylinder**, producing power by expanding gases that push a piston or rotate a turbine.

Types of IC Engines

1. Based on Fuel Used:

- Petrol (Gasoline) Engine – Spark Ignition (SI) engine
- Diesel Engine – Compression Ignition (CI) engine
- Gas Engines – Use CNG, LPG, etc.

2. Based on Number of Strokes:

- 2-Stroke Engine – Power stroke every revolution
- 4-Stroke Engine – Power stroke every two revolutions

3. Based on Applications:

- Automotive Engines
- Marine Engines
- Aircraft Engines
- Stationary Engines (for pumps/generators)

1.13 2 STROKE and 4 STROKE ENGINES

1.3.1. 2 STROKE ENGINES

A Two-stroke Engine is a type of internal combustion engine that completes one rotation of the crankshaft in two motions of the piston. It completes the entire cycle of rotation in one stroke of the piston.

The different parts are listed as follows:

Piston: The piston moves by the forces generated by the combustion of the gasses. The piston converts this chemical energy into mechanical energy for the crankshaft.

Crankshaft: It is connected to the pistons and converts the reciprocal motion of the pistons into the rotational motion which drives the driving gears.

Connecting Rod: It connects the piston head to the crankshaft.

Inlet Port: It is the opening for fuel and air mixture to come inside the piston chamber.

Exhaust Port: It is the opening for the exhaust fumes which opens during the downstroke.

Spark Plug: Spark plug is the device inside the piston chamber that causes the combustion of the fuel-air mixture.

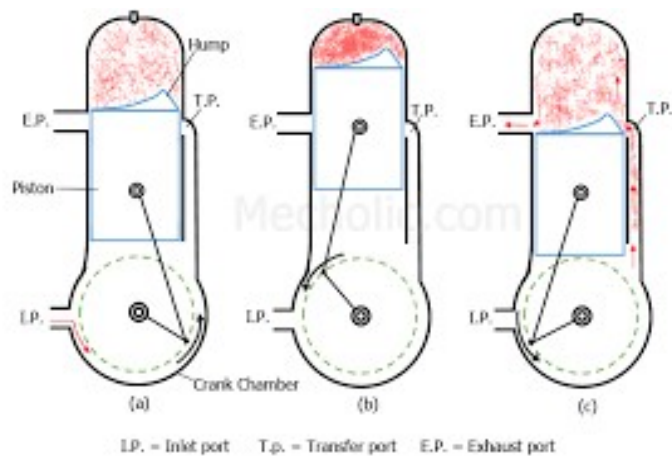


Fig 1.9 2 Stroke Engines

1.13.2. 4 STROKE ENGINES

- The Four-Stroke Spark Ignition (SI) Engine
- A four-stroke SI engine is a type of internal combustion engine that operates in a cycle of four strokes: intake, compression, power, and exhaust. Each stroke represents a movement of the piston up or down within the cylinder.

Working Principle

Intake Stroke:

- The intake valve opens as the piston moves downward, creating a vacuum.
- A mixture of air and fuel is drawn into the cylinder.

Compression Stroke:

- Both valves are closed, and the piston moves upward, compressing the air-fuel mixture.

- This compression increases the temperature and pressure of the mixture.

Power Stroke:

- A spark plug ignites the compressed mixture, causing a powerful explosion that forces the piston downward.
- This downward motion converts the energy of the explosion into mechanical work.

○

Exhaust Stroke:

- The exhaust valve opens, and the piston moves upward, forcing the burnt gases out of the cylinder.
- The engine is now ready for another cycle.

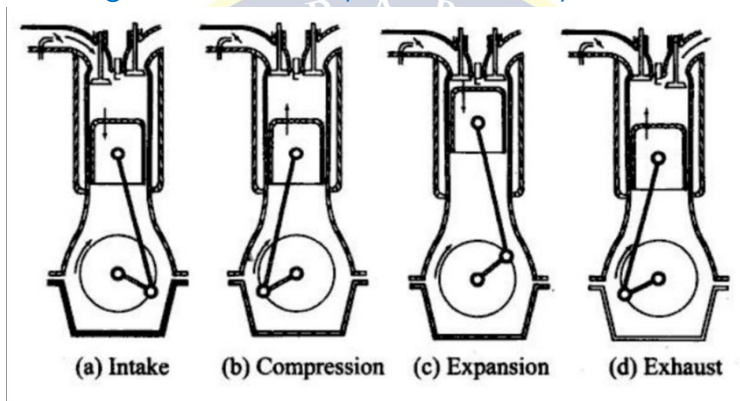


Fig 1.10 4 Stroke Engines

Key components:

- Cylinder: The chamber where the combustion process takes place.
- Piston: A movable component that reciprocates within the cylinder.
- Crankshaft: Converts the linear motion of the piston into rotary motion.
- Connecting rod: Connects the piston to the crankshaft.
- Intake valve: Allows air and fuel to enter the cylinder.
- Exhaust valve: Allows burnt gases to exit the cylinder.
- Spark plug: Ignites the air-fuel mixture.

1.14. SI/CI Engines

Feature	CI Engines	SI Engines
Ignition	Compression of the air-fuel mixture causes spontaneous ignition.	A spark plug ignites the air-fuel mixture.
Fuel	Diesel fuel	Gasoline or petrol
Compression Ratio	Higher compression ratio (typically 15-25)	Lower compression ratio (typically 8-12)
Fuel Injection	Direct fuel injection into the combustion chamber.	Indirect fuel injection into the intake manifold.
Efficiency	Generally more efficient than SI engines, especially at higher loads.	Less efficient than CI engines, especially at lower loads.
Emissions	Typically produce lower emissions of particulate matter but higher emissions of nitrogen oxides.	Typically produce lower emissions of nitrogen oxides but higher emissions of particulate matter.
Noise	Generally quieter than SI engines.	Generally noisier than CI engines.
Applications	Heavy-duty vehicles, such as trucks, buses, and construction equipment.	Light-duty vehicles, such as cars and motorcycles.

1.15. COMPONENTS OF ELECTRIC VEHICLES

Main Components of Electric Vehicles

1. Battery Pack

- Stores electrical energy used to power the electric motor.
- Common types: Lithium-ion, Nickel-metal hydride, Solid-state (emerging).
- Key specs: Voltage, Capacity (kWh), Range, Charging time.

2. Electric Motor

- Converts electrical energy from the battery into mechanical energy to drive the wheels.

- Types of motors:
 - DC Motor
 - Brushless DC Motor (BLDC)
 - Induction Motor (AC)
 - Permanent Magnet Synchronous Motor (PMSM)

3. **Power Electronics Controller**

- Manages the flow of electrical energy between the battery and the motor.
- Controls motor speed, torque, and regenerative braking.
- Includes inverter (DC to AC conversion for AC motors).

4. **Onboard Charger**

- Converts AC power from the grid into DC power to charge the battery.
- Handles communication between the charging station and vehicle.

5. **Transmission / Drivetrain**

- Simplified compared to internal combustion vehicles.
- May include:
 - Single-speed gearbox
 - Reduction gear
 - Direct drive (in some EVs)

6. **Regenerative Braking System**

- Captures kinetic energy during braking and converts it into electrical energy.
- Stores this energy back into the battery, improving efficiency.

7. **Vehicle Control Unit (VCU) / Central Controller**

- Coordinates operation of all subsystems.
- Ensures safe and efficient functioning of battery, motor, charging, etc.

8. **Battery Management System (BMS)**

- Monitors battery temperature, voltage, current, and state of charge (SOC).
- Prevents overcharging, overheating, and battery damage.

9. **Thermal Management System**

- Maintains optimal operating temperature for battery, motor, and electronics.
- Includes cooling fans, liquid coolant loops, or heat pumps.

10. ⚙️ Auxiliary Systems

- Power steering, HVAC (Heating, Ventilation, and Air Conditioning), infotainment, lighting—usually powered by a 12V system from the main battery.

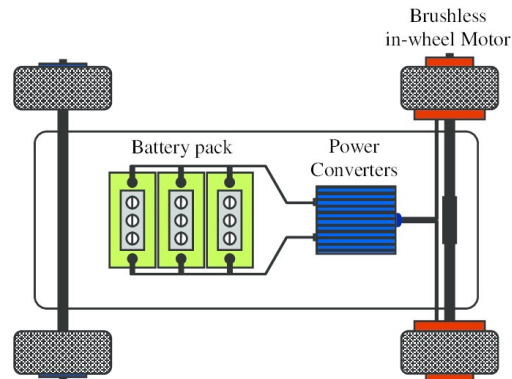


Fig 1.11 Electric Vehicle

1.16. COMPONENTS OF HYBRID VEHICLES

Hybrid Vehicles: A Blend of Power

- Hybrid vehicles combine the power of a traditional internal combustion engine (ICE) with an electric motor, offering improved fuel efficiency, reduced emissions, and a quieter driving experience.
- Components of a Hybrid Vehicle
- Internal Combustion Engine (ICE): Provides power through traditional combustion processes.
- Electric Motor: Offers additional power and can operate independently.
- Battery Pack: Stores energy for the electric motor.
- Power Electronics Control Unit (PECU): Manages the flow of power between the ICE, electric motor, and battery.
- Transmission: Controls the power transfer from the ICE and electric motor to the wheels.
- Generator: Converts mechanical energy from the ICE into electrical energy to charge the battery.
- Regenerative Braking System: Captures kinetic energy during braking and stores it in the battery.

- Working Principle of a Hybrid Vehicle
- Hybrid vehicles operate in various modes, depending on driving conditions and the driver's input:
- Electric Mode: The electric motor powers the vehicle, using energy stored in the battery. This mode is ideal for low-speed driving and short distances.
- Hybrid Mode: The ICE and electric motor work together to provide power. The ICE may be used for acceleration or higher speeds, while the electric motor assists during peak power demands.
- Engine Mode: The ICE powers the vehicle directly, similar to a traditional gasoline or diesel car. This mode is typically used for highway driving or when the battery's charge is low.
- Charging Mode: The vehicle can be plugged into an external power source to charge the battery.

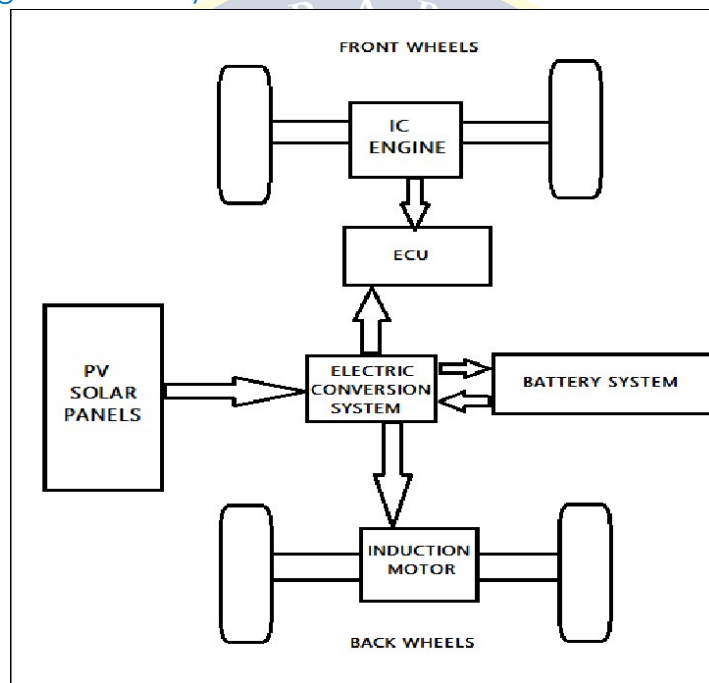


Fig 1.12 Hybrid Vehicle

9. Practice Quiz

1. Sprue in casting refers to

- a) Runner
- b) Vertical Passage
- c) Raiser

d) Gate

2. Which of the following metal forming process performs squeezing out of material through a hole

a) Forging

b) Drawing

c) Extrusion

d) Rolling

3. Which type of transformers is used in AC Welding

a) Equal turns ratio

b) Ferrite Core

c) Step Up

d) Step Down

4. What makes 3D Printing superior to other crafting techniques

a) Efficiency

b) Customization

c) Quality

d) Speed

5. A four stroke petrol engine theoretically operates on _____ cycle

a) **Otto**

b) Brayton

c) Joule

d) Bell coleman

6. A Two-stroke engine gives _____ mechanical efficiency than a four stroke engine

a) **higher**

b) lower

c) equal

d) none

7. Hydro electric power plant is mainly located in _____ areas

a) **Hilly**

b) Flat

c) Delats

d) Deserts

8. The first Nuclear reactor was set up in India at _____

a) Bombay

b) Rana Pratap Sagar

c) Kalpakkam

d) Tarapur

10. Assignments

S.No	Question	BL	CO
1	What is mean by casting? explain the principles of casting with neat sketch?	2	2
2	Distinguish in between Conventional Machining and CNC Machining	2	2
3	Explain various components and working of Hybrid vehicles with neat sketch.	2	2
4	Explain the working principle of Four stroke S.I engine with a neat sketch	2	2
5	What do you mean by Refrigeration and Airconditioning and explain the working principle of refrigeration system with neat sketch?	2	2

11. Part A- Question & Answers

S.No	Question& Answers	BL	CO
1	Define casting in manufacturing process. Ans. The casting process is a manufacturing method that produces desired-shape metal parts by pouring molten metal into a mold and solidifying and cooling it to the room temperature	1	2
2	Write various components in hybrid vehicles. Ans. The hybrid car has five main components: the gas engine, the electric motor, the battery, the power control unit, and the regenerative braking system.	1	2
3	Write a short note on Welding process. Ans. Welding is a fabrication process that join materials like metals by using heat at high temperatures.	1	2
4	What is the main purpose of Boiler Ans. The main purpose of boiler is to produce steam from water.	1	2
5	Define the term refrigeration Ans. The word refrigeration refers to the process of cooling space, material, or system to lower and maintain its temperature under room temperature.	1	2

12. Part B- Questions

S.No	Question	BL	CO
1	What is mean by casting? explain the principles of casting with neat sketch?	1	2
2	What is mean by Joining process and explain the types of Joining process?	1	2
3	Distinguish between Conventional Machining and CNC Machining	1	2
4	Explain various components and working of Hybrid vehicles with neat sketch.	2	2
5	Explain the working principle of Four stroke S.I engine with a neat sketch	2	2

13. Supportive Online Certification Courses

1. Advanced materials and processes By Prof. Jayanta Das, conducted by IIT Kharagpur- 12 weeks
2. Computer Integrated Manufacturing By Prof. J.Ramkumar, conducted by IIT Kanpur – 12 weeks.

14. Real Time Applications

S.No	Application	CO
1	Casting Application: Manufacturing of engine blocks, turbine blades, and complex metal parts in automotive and aerospace industries. Real-time Example: Aluminum cylinder heads cast in sand molds for car engines.	1
2	Joining Processes (Welding, Brazing, Soldering, Adhesive Bonding) Application: Shipbuilding, pipeline construction, aerospace assembly, and automotive chassis joining. Real-time Example: Spot welding in automotive manufacturing for body panels.	1
3	IC Engines (Internal Combustion Engines) Application: Automobiles, power backup systems, marine vessels. Real-time Example: Engines in Royal Enfield bikes or Tata diesel trucks.	1
4	Introduction to CNC Machines Application: Mass production of precision components with high repeatability.	1

	Real-time Example: CNC lathes and mills in the aerospace sector for producing engine parts.	
5	3D Printing (Additive Manufacturing) Application: Rapid prototyping, dental implants, customized consumer products, lightweight aerospace parts. Real-time Example: 3D-printed jet engine brackets in GE's LEAP engine.	1

15. Contents Beyond the Syllabus

1. 3D Milling

3d machining typically used STL files from 3D CAD programs as a source and requires all three axes of the mill to be able to move simultaneously. 3D machining can take a very long time even on fast, modern machines and especially if a high quality finish is required. A fourth or rotary axis can also be added to create turned parts.

16. Prescribed Text Books & Reference Books

Text Book

1. R.S. Khurmi & J.K. Gupta – “A Textbook of Basic Mechanical Engineering” – S. Chand Publications
2. Sharma & Purohit – “Basic Mechanical Engineering” – Pearson

References:

1. P.L. Ballaney – “Thermal Engineering” – Khanna Publishers

17. Mini Project Suggestion

1.Low-Cost 3D Printer Prototype using Recycled Parts

Objective:

Design and develop a small-scale, low-cost 3D printer using recycled parts from old printers/scanners and open-source electronics like Arduino.

Key Concepts Covered:

Additive manufacturing principles
 CNC machine axis movement (X, Y, Z control)
 Stepper motor control
 Smart manufacturing integration using sensors (temperature, end-stop)

Real-time Application:

Affordable rapid prototyping for small startups, schools, or hobbyists.

Tools/Materials Needed:

- Arduino Uno + RAMPS board
- Stepper motors from old printers
- Hotend + heated bed